# Experience-Curve Analysis: Silk Road Salts

## Public Comparable: COMPASS MINERALS INTERNATIONAL INC (CMP)

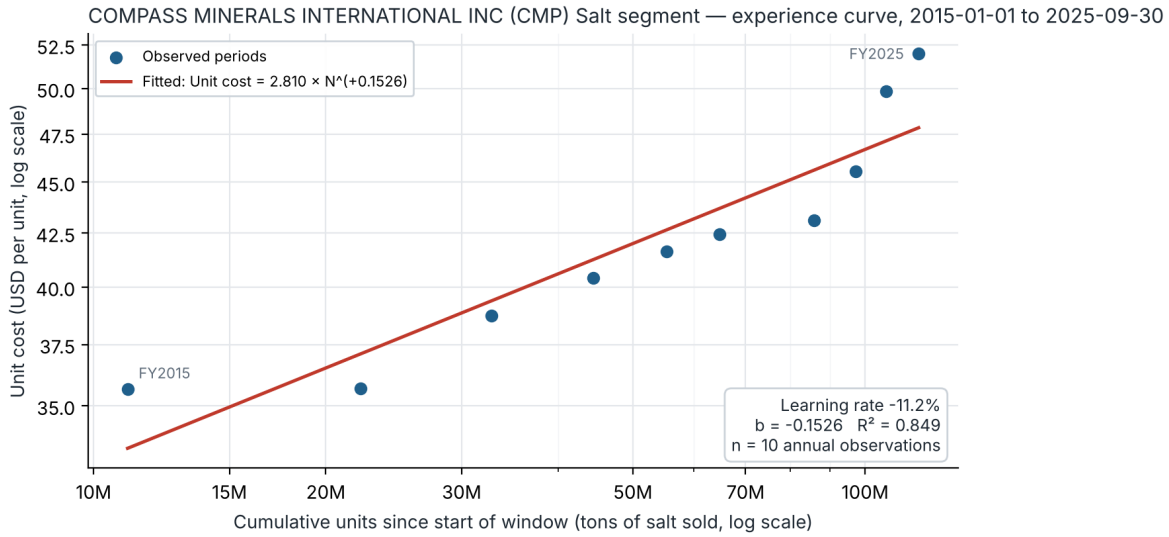
**Tool:** Learning-Curve Calculator
**Public Tool URL:** srslearningcurve.streamlit.app

## Learning-Curve Results

| | |
|---|---|
| **Company evaluated** | Silk Road Salts |
| **Public comparable** | COMPASS MINERALS INTERNATIONAL INC (CMP), SEC CIK 1227654 — Salt segment |
| **Analysis period** | FY2015 – FY2025 (2015-01-01 to 2025-09-30) |
| **Data frequency** | Annual |
| **Learning rate** | -11.2% |
| **Coefficient b** | -0.1526 (standard error 0.0227) |
| **R² (log-log)** | 0.849 |
| **Observations** | 10 |
| **Regression equation** | Unit cost = 2.810 × N^(+0.1526), where N = cumulative tons of salt sold |
| **Unit-cost methodology** | Unit cost = period segment cost ÷ period tons of salt sold (USD per unit) |
| **Unit measurement** | Direct segment unit volumes reported in SEC filings (each figure cited) |
| **Cost measure** | Salt segment operating cost excluding shipping and handling (derived: Salt sales − shipping and handling cost − Salt operating earnings) |

### Why this comparable

McCormick & Company (MKC), the closer strategic comparable, was evaluated first and does not meet the data requirements: its fiscal 2020–2025 10-Ks disclose no absolute unit volumes (only a combined percentage 'volume and product mix' change), report Consumer-segment cost of goods sold only for fiscal 2023–2025, and give no documented average selling price for a revenue-based proxy. Compass Minerals reports Salt-segment tons sold and the segment figures needed to measure Salt-segment cost in every year, so costs and volumes cover the same business activity.



### Interpretation

Across 10 annual observations (2015-01-01 to 2025-09-30), COMPASS MINERALS INTERNATIONAL INC's Salt segment unit cost rose by about 11.2% for each doubling of cumulative units (b = -0.1526), the opposite of an experience-curve effect. The log-log fit is strong ($R^2$ = 0.849), meaning costs rose consistently as volume accumulated; no experience curve is visible. On this evidence, cumulative volume alone should not be treated as a durable source of cost advantage, and the company's position relative to its industry cannot be determined without a verified benchmark. Key limitations: costs are nominal (not inflation-adjusted), cumulative units are counted only from the start of the selected window, and because volume per period was fairly stable, cumulative volume rose almost in step with time, so the fit cannot separate volume effects from cost changes over time.

Interpretation is analysis generated from the computed results, not a reported fact.

### Qualifications

- TP2021 (9 mo) covers 273 days, not a full annual period: its units count toward cumulative volume, but it is excluded from the regression.

## Sources and Strategic Implications

Reported = figure as stated in the cited filing. Derived = arithmetic on reported figures, shown in full. Calculated = computed by this tool (cumulative units, unit cost, regression).

### Period-level calculation

| Period | Start | End | Segment cost (USD m) | Units | Cumulative units | Unit cost (USD) | Used |
|---|---|---|---|---|---|---|---|
| FY2015 | 2015-01-01 | 2015-12-31 | 394.7 | 11,069,000 | 11,069,000 | 35.66 | yes |
| FY2016 | 2016-01-01 | 2016-12-31 | 396.8 | 11,113,000 | 22,182,000 | 35.71 | yes |
| FY2017 | 2017-01-01 | 2017-12-31 | 410.6 | 10,600,000 | 32,782,000 | 38.74 | yes |
| FY2018 | 2018-01-01 | 2018-12-31 | 470.0 | 11,627,000 | 44,409,000 | 40.42 | yes |
| FY2019 | 2019-01-01 | 2019-12-31 | 454.9 | 10,923,000 | 55,332,000 | 41.65 | yes |
| FY2020 | 2020-01-01 | 2020-12-31 | 400.6 | 9,440,000 | 64,772,000 | 42.44 | yes |
| TP2021 (9 mo) | 2021-01-01 | 2021-09-30 | 339.1 | 8,510,000 | 73,282,000 | 39.85 | no |
| FY2022 | 2021-10-01 | 2022-09-30 | 542.2 | 12,575,000 | 85,857,000 | 43.12 | yes |
| FY2023 | 2022-10-01 | 2023-09-30 | 515.8 | 11,320,000 | 97,177,000 | 45.57 | yes |
| FY2024 | 2023-10-01 | 2024-09-30 | 464.1 | 9,314,000 | 106,491,000 | 49.83 | yes |
| FY2025 | 2024-10-01 | 2025-09-30 | 564.1 | 10,848,000 | 117,339,000 | 52.00 | yes |

Used = no for TP2021 (9 mo): Nine-month transition period (fiscal year-end changed from December 31 to September 30); not comparable with full-year unit costs. Its units are still counted in cumulative units.

### SEC sources for segment cost (Salt segment)

| Period | Value (USD) | Form | Filed | Accession (link) | How obtained |
|---|---|---|---|---|---|
| FY2015 | 394,700,000 | 10-K | 2018-02-27 | 0001227654-18-000031 | Derived: Salt sales $849.0M − shipping and handling cost $239.1M − Salt operating earnings $215.2M = $394.7M (Note 14, Operating Segments) |
| FY2016 | 396,800,000 | 10-K | 2019-03-01 | 0001227654-19-000037 | Derived: Salt sales $811.9M − shipping and handling cost $214.5M − Salt operating earnings $200.6M = $396.8M (Note 15, Operating Segments) |
| FY2017 | 410,600,000 | 10-K | 2020-02-26 | 0001227654-20-000044 | Derived: Salt sales $769.2M − shipping and handling cost $220.6M − Salt operating earnings $138.0M = $410.6M (Note 15, Operating Segments) |
| FY2018 | 470,000,000 | 10-K | 2021-02-26 | 0001227654-21-000037 | Derived: Salt sales $858.1M − shipping and handling cost $272.4M − Salt operating earnings $115.7M = $470.0M (Note 15, Operating Segments) |
| FY2019 | 454,900,000 | 10-KT | 2021-11-30 | 0001227654-21-000309 | Derived: Salt sales $889.5M − shipping and handling cost $267.4M − Salt operating earnings $167.2M = $454.9M (Note 13, Operating Segments) |

| Period | Value (USD) | Form | Filed | Accession (link) | How obtained |
|---|---|---|---|---|---|
| FY2020 | 400,600,000 | 10-K | 2022-12-14 | 0001227654-22-000340 | Derived: Salt sales $779.4M − shipping and handling cost $217.8M − Salt operating earnings $161.0M = $400.6M (Note 14, Operating Segments) |
| TP2021 (9 mo) | 339,100,000 | 10-K | 2023-11-29 | 0001227654-23-000226 | Derived: Salt sales $671.1M − shipping and handling cost $198.8M − Salt operating earnings $133.2M = $339.1M (Note 15, Operating Segments) |
| FY2022 | 542,200,000 | 10-K | 2024-12-16 | 0001227654-24-000230 | Derived: Salt sales $1,011.4M − shipping and handling cost $353.6M − Salt operating earnings $115.6M = $542.2M (Note 14, Operating Segments) |
| FY2023 | 515,800,000 | 10-K | 2025-12-12 | 0001227654-25-000199 | Derived: Salt sales $1,010.8M − shipping and handling cost $324.5M − Salt operating income $170.5M = $515.8M (Note 12, Operating Segments) |
| FY2024 | 464,100,000 | 10-K | 2025-12-12 | 0001227654-25-000199 | Derived: Salt sales $907.8M − shipping and handling cost $280.1M − Salt operating income $163.6M = $464.1M (Note 12, Operating Segments) |
| FY2025 | 564,100,000 | 10-K | 2025-12-12 | 0001227654-25-000199 | Derived: Salt sales $1,022.5M − shipping and handling cost $312.5M − Salt operating income $145.9M = $564.1M (Note 12, Operating Segments) |

## Unit volume sources

Method: Direct segment unit volumes reported in SEC filings (each figure cited). Unit definition: tons of salt sold. Scope confirmed to match the segment cost: yes. The filings report tons sold in thousands; values here are tons (thousands × 1,000).

| Period | Units | Source |
|---|---|---|
| FY2015 | 11,069,000 | 10-K 0001227654-18-000031, Item 7 MD&A, Salt Segment Results table: total tons sold 11,069 thousand |
| FY2016 | 11,113,000 | 10-K 0001227654-19-000037, Item 7 MD&A, Salt Segment Results table: total tons sold 11,113 thousand |
| FY2017 | 10,600,000 | 10-K 0001227654-20-000044, Item 7 MD&A, Salt Segment Results table: total tons sold 10,600 thousand |
| FY2018 | 11,627,000 | 10-K 0001227654-21-000037, Item 7 MD&A, Salt Segment Results table: total tons sold 11,627 thousand |
| FY2019 | 10,923,000 | 10-KT 0001227654-21-000309, Item 7 MD&A, Salt Segment Results table: total tons sold 10,923 thousand |
| FY2020 | 9,440,000 | 10-KT 0001227654-21-000309, Item 7 MD&A, Salt Segment Results table: total tons sold 9,440 thousand |
| TP2021 (9 mo) | 8,510,000 | 10-K 0001227654-22-000340, Item 7 MD&A, Salt Segment Results table: total tons sold 8,510 thousand |
| FY2022 | 12,575,000 | 10-K 0001227654-24-000230, Item 7 MD&A, Salt Segment Results table: total tons sold 12,575 thousand |
| FY2023 | 11,320,000 | 10-K 0001227654-24-000230, Item 7 MD&A, Salt Segment Results table: total tons sold 11,320 thousand |
| FY2024 | 9,314,000 | 10-K 0001227654-25-000199, Item 7 MD&A, Salt Segment Results table: total tons sold 9,314 thousand |
| FY2025 | 10,848,000 | 10-K 0001227654-25-000199, Item 7 MD&A, Salt Segment Results table: total tons sold 10,848 thousand |

## Methodology

Cost measure: Salt segment operating cost excluding shipping and handling (derived: Salt sales − shipping and handling cost − Salt operating earnings). Segment figures are not published in the SEC Company Facts API (it holds only consolidated facts), so each value was transcribed from the segment note or MD&A of the cited 10-K/10-KT (CIK 1227654); the table above gives the form, filing date, accession number and derivation for every figure. Costs and units are for the same segment.

When a period appears in several filings, the most recently filed figure is used so that restatements are reflected; differences from earlier filings are listed under data notes.

Unit volumes: Direct segment unit volumes reported in SEC filings (each figure cited). Unit definition: tons of salt sold.

Cumulative units N(t) = sum of units from the first period in the window through period t. Unit cost C(t) = period segment cost ÷ period units. The model C = A·N^(−b) is fitted by ordinary least squares of ln C on ln N. Learning rate = 1 − 2^(−b); $R^2$ is from the log-space regression. Periods that are not full fiscal years (TP2021 (9 mo)) add their units to N but are excluded from the regression, so cumulative volume has no gap and every regression observation covers a full year.

### Assumptions and limitations

- Cumulative production is measured from the beginning of the selected data window, not from the company's founding. True lifetime cumulative production is generally unavailable.
- The regression estimates an association between cumulative units and unit cost; it is not proof of a causal learning effect. Input prices, inflation, product mix, accounting choices and capacity utilization can all move unit costs.
- Segment cost is in nominal currency; inflation and input-price changes are not removed.
- Costs and units both come from the same segment's disclosures; consolidated company COGS was not used.
- The cost measure is Salt product cost plus Salt-segment SG&A and other operating items, excluding shipping and handling; where the FY2025 10-K discloses the split (fiscal 2023–2025), SG&A and other items are 5.5–7.0% of the measure.
- Segment operating results include occasional disclosed special items that were not removed (for example restructuring charges of $4.3M in 2017 and $5.5M in fiscal 2023 and $2.8M of 2019 flood-related logistics costs; the filings do not always state which segment bore them), and the FY2018 10-K attributes higher Salt per-unit costs to a 2017 ceiling fall and an April–July 2018 strike at the Goderich mine.
- Highway-deicing tons swing with winter weather, and the mix between deicing salt and higher-priced consumer and industrial salt shifts cost per ton for reasons unrelated to learning.
- Compass already sold 11.1 million tons of salt in 2015, the first year of the window, so its true cumulative production is far larger than the cumulative tons measured here and the window contains far fewer genuine doublings of experience.

### What this means for Silk Road Salts

COMPASS MINERALS INTERNATIONAL INC (Salt segment) does not show evidence of production-related cost advantages over this window: unit cost rose about 11.2% per doubling of cumulative volume ($R^2$ = 0.849), so the data do not establish a meaningful experience-curve advantage. Accumulated volume therefore does not appear to give the comparable a learning-driven cost lead that would by itself shut out smaller competitors; any scale advantage would have to come from other sources (such as fixed-cost absorption, purchasing or logistics), which this analysis does not measure. The measured rate should not be assumed to transfer to Silk Road Salts. Compass Minerals' Salt segment is dominated by bulk highway deicing salt (8,985 of 10,848 thousand tons in fiscal 2025, averaging $71.53 per ton, versus $203.87 per ton for consumer and industrial salt), mined or evaporated at very large scale, where throughput, energy, mine conditions and logistics drive costs. Silk Road Salts sells premium culinary and finishing salts and seasoning blends at far smaller scale, where sourcing, small-batch processing, packaging and brand building are likely to matter more than cumulative production volume. This analysis does not predict the learning rate Silk Road Salts will achieve.

Interpretation, not a reported fact: the comparable is an imperfect industry reference.

### Data notes

- FY2019: Salt operating earnings were $168.0M in the FY2019 and FY2020 10-Ks; the later 10-KT recast them to $167.2M (used).
- FY2020: Salt operating earnings were $161.8M in the FY2020 10-K; the 10-KT and FY2022 10-K report $161.0M (used).
- FY2022: The FY2022 and FY2023 10-Ks reported Salt sales $1,010.3M, shipping and handling $353.3M, operating earnings $116.2M (cost $540.8M) and 12,557 thousand tons; the FY2024 10-K figures (used) are

$1,011.4M, $353.6M, $115.6M and 12,575 thousand tons.

- FY2023: Salt operating earnings were $170.7M in the FY2023 10-K; the FY2024 and FY2025 10-Ks report $170.5M (used). The FY2025 10-K also discloses Salt product cost $481.3M and segment SG&A $34.5M, which sum to the derived $515.8M.
- FY2024: Cross-check: the FY2025 10-K discloses Salt product cost $431.8M, SG&A $31.3M and other operating expense $1.0M, which sum to the derived $464.1M.
- FY2025: Cross-check: the FY2025 10-K discloses Salt product cost $533.0M, SG&A $30.5M and other operating expense $0.6M, which sum to the derived $564.1M.

### Automatic checks passed

- No duplicated reporting periods; no gaps between consecutive periods in the window.
- Every regression period has a duration consistent with annual reporting; TP2021 (9 mo) counts toward cumulative units only.
- Every period has positive units with a documented source; any period with non-positive segment cost is excluded from the regression but its units still count toward cumulative volume.
- At least 5 usable observations (n = 10).

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